

Midterm Examination #3

Indicate (by circling or underlining carefully) *one and only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

- Which of the gas laws states the volume of a gas is directly proportional to amount of a gas in moles ($V=k \times n$)?
(a) Boyle's Law (b) Charles' Law (c) Gay-Lussac's Law
(d) Avogadro's Law (e) Ideal Gas Law
- In a redox reaction, the species that **loses electrons** is said to be:
(a) reduced (b) precipitated (c) oxidized (d) dissolved (e) unchanged
- A chemical reaction where two or more simple reactants combine to form a single, more complex product (e.g., $A + B \rightarrow AB$) is what type of reaction?
(a) decomposition (b) single displacement (SD) (c) double displacement (DD)
(d) combination (synthesis) reaction (e) combustion reaction
- What is molarity of aqueous solution in which 0.50 moles KCl dissolved to final 500.0 mL volume?
(a) 0.25 M (b) 0.50 M (c) 1.0 M (d) 2.0 M (e) 0.0010 M
- If you take 10.0 mL of a 2.5 M stock solution of NaCl and dilute it with water to a final volume of 50.0 mL, what is the final concentration (C_2)?
(a) 0.10 M (b) 0.50 M (c) 1.25 M (d) 5.0 M (e) 0.050 M
- A solution is prepared by mixing 10.0 g of sugar (solute) with 90.0 g of water (solvent). What is the percent by mass of the sugar in the solution?
(a) 1.0% (b) 10.0% (c) 11.1% (d) 90.0% (e) 100.0%
- Calculate the **pressure** of $n = 2.00$ mol of gas at $T = 273$ K, $V = 44.8$ L. ($R = 0.0821$ L-atm/[mol·K])
(a) 1.00 atm (b) 2.00 atm (c) 0.500 atm (d) 22.4 atm (e) none of the above
- Convert a **pressure** of 760 mmHg to atm.
(a) 0.500 atm (b) 1.00 atm (c) 7.60 atm (d) 76.0 atm
- Which law states that **pressure** and **volume** are inversely related, that is $P_1V_1 = P_2V_2$?
(a) Boyle's Law (b) Charles' Law (c) Gay-Lussac's Law (d) Avogadro's Law
- Which of the following is a **combustion** reaction?
(a) $CH_4 + 2 O_2 \rightarrow CO_2 + 2 H_2O$ (b) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
(c) $HCl + NaOH \rightarrow NaCl + H_2O$ (d) $CaCO_3 \rightarrow CaO + CO_2$
(e) none of the above
- Which of the following is the correct oxidation number of **hydrogen** in H_2O molecule?
(a) 0 (b) -1 (c) +1 (d) +2 (e) none of the above
- A substance that completely dissociates into ions when dissolved in an aqueous solution and conducts electricity strongly is classified as a:
(a) non-electrolyte (b) weak electrolyte (c) strong electrolyte
(d) saturated solution (e) suspension

13. What is the oxidation number (ON) for a neutral atom of Cl in its elemental state, Cl_2 ?
 (a) -1 (b) 0 (c) +1 (d) +2 (e) +7
14. The strongest type of intermolecular force that exists between two molecules of H_2O (water) is:
 (a) dispersion forces (London) (b) dipole-dipole forces (c) ion-dipole forces
 (d) Hydrogen bonding (e) covalent bonding
15. How many moles are in 15.0 g of KOH? (molar mass of KOH = 56.11 g/mol)
 (a) 0.188 mol (b) 0.267 mol (c) 0.459 mol (d) 1.25 mol
16. Which is a key assumption of the **Kinetic Molecular Theory** for an *ideal gas*?
 (a) gas molecules exert significant attractive forces on each other
 (b) the molecules that make up the gas phase must interact with their liquid phase
 (c) gas collisions are completely elastic (energy is not lost in collision)
 (d) the pressure of the gas is inversely proportional to temperature
17. Which of the following salts is generally considered **soluble** in water?
 (a) AgCl (b) PbI_2 (c) BaSO_4 (d) KNO_3 (e) CaCO_3
18. A solution that contains the maximum amount of dissolved solute possible at a given temperature is called a:
 (a) unsaturated solution (b) saturated solution (c) supersaturated solution
 (d) dilute solution (e) concentrated solution
19. A gas has $P_1 = 2.00$ atm, $V_1 = 4.00$ L. If pressure increases to $P_2 = 8.00$ atm, what is V_2 ?
 (a) 0.50 L (b) 1.00 L (c) 2.00 L (d) 16.0 L (e) none of the above
20. Hydrogen bonding can only occur between molecules where hydrogen is directly bonded to one of three highly electronegative atoms. These atoms are:
 (a) C, H, O (b) S, Cl, Br (c) Cl, Br, I (d) N, O, F (e) Na, K, Li
21. In a sugar-water solution, the sugar is the **solute** and the water is the **solvent**. The **solvent** is defined as the component present in:
 (a) the smallest amount (b) the solid phase (c) the greatest amount
 (d) the gaseous phase (e) the ionic form
22. The concentration unit of **molarity (M)** is correctly expressed in which of the following units?
 (a) mol/kg (b) g/L (c) mol/mL (d) mol/L (e) kg/mol
23. Which of the following is a spectator ion in the reaction:
 $\text{NaCl (aq)} + \text{AgNO}_3 \text{ (aq)} \rightarrow \text{AgCl (s)} + \text{NaNO}_3 \text{ (aq)}$
 (a) Na^+ (b) Cl^- (c) Ag^+ (d) NO_3^- (e) both (a) and (d)
24. Which statement best describes "like dissolves like"?
 (a) polar solutes dissolve in polar solvents (b) nonpolar solutes dissolve in nonpolar solvents
 (c) ionic solutes dissolve in polar solvents (d) all of the above
 (e) none of the above
25. A precipitate forms and you have to write the chemical reaction equation. How will you indicate the compound that precipitated?
 (a) (s) (b) (l) (c) (g) (d) (aq) (e) none of these

Circle either **True** or **False** for the following statements:

26. A solution that conducts electricity must contain ions.

True ☒ False

27. Combustion reactions always involve oxygen as a product.

True ☒ False

28. In a balanced chemical equation, the number and type of atoms must be equal on both sides.

True ☒ False

29. The oxidation number of a pure element is always zero.

True ☒ False

30. In a chemical reaction equation, compounds to the right of the reaction arrow are called **reactants**.

True ☒ False

31. The oxidation number of any Group 1 metal (like Na or K) in any compound is always +1.

True ☒ False

32. The total pressure of a gas mixture is equal to the sum of the partial pressures of all the individual gases in the mixture.

True ☒ False

33. The solid that forms from a solution during a chemical reaction, often by mixing two solutions, is called a **precipitate**.

True ☒ False

34. At STP, the temperature is 25 °C and pressure is 1 atm.

True ☒ False

35. The variable R in the Ideal Gas Law equation ($PV=nRT$) is known as the **universal gas constant**.

True ☒ False

Midterm Examination #1

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- Which element belongs to the **alkali metals** group?
(a) calcium (b) sodium (c) **magnesium** (d) chlorine
- Which subatomic particle has a mass of approximately **1 amu** and a positive charge?
(a) electron (b) **proton** (c) neutron (d) none of the above
- Which of the following is a **polyatomic ion**?
(a) Cl^- (b) Na^+ (c) **SO_4^{2-}** (d) O (e) none of the above
- Convert 25.0°C into Kelvin.
(a) 248 K (b) 273 K (c) **298 K** (d) 300 K
- A sample has a mass of 12.5 g and a volume of 2.50 mL. What is its density?
(a) 0.200 g/mL (b) **5.00 g/mL** (c) 31.3 g/mL (d) 0.050 g/mL
- Which of the following is the correct name for $\text{Ca}(\text{NO}_3)_2$?
(a) cobalt fluoride (b) calcium nitroxide (c) **calcium nitrate**
(d) calcium oxide (e) kryptonite
- Which of the following terms describes the distance between two wave crests?
(a) wave speed (b) frequency (c) **wavelength** (d) amplitude
- Which of the following is the correct atomic notation for carbon-14?
(a) $^{12}_6\text{C}$ (b) **$^{14}_6\text{C}$** (c) $^{14}_6\text{C}$ (d) $^{60}_{27}\text{Co}$
- Convert 4.56×10^{-3} into standard notation
(a) **0.00456** (b) 4560 (c) 0.000456 (d) 4,560
- Which of the following is the electron configuration for hydrogen?
(a) **$1s^1$** (b) $1s^2$ (c) $1s^2 2s^1$ (d) $1p^1$ (e) $1s^2 2p^1$
- Which of the following is true about element bromine (Br)?
(a) It is a member of the p-block of elements
(b) It is a Group 17 element
(c) It is a non-metal
(d) **all of the above**
(e) none of the above
- What term describes an electron normally in $n = 1$ shell acquiring energy to be in $n = 2$ shell?
(a) emission (b) **excited state** (c) desorption (d) ground state (e) nucleated
- Express 0.0000721 in scientific notation with correct significant figures.
(a) 7.21×10^{-4} (b) **7.21×10^{-5}** (c) 7.21×10^{-6} (d) 7.21×10^{-7} (e) none of the above
- When you complete the operation on the expression $1.450 \times 3.01 \div 10.025$, how many significant digits will the final result have?
(a) 2 (b) 3 (c) **4** (d) 5 (e) 7

15. The scientific method includes a stage where a **tentative explanation** is given to account for a set of related observations. What is that stage?
(a) law ☒ (b) hypothesis (c) theory (d) experimentation (e) the Atomic Model
16. Which of these is a polyatomic ion?
(a) Ne (b) H^+ ☒ (c) CO_3^{2-} (d) Br (e) Na^+
17. Which of these is the ground-state electron configuration for the **carbon (C)** atom?
(a) $1s^2 2s^2 2p^5$ (b) $1s^2 2s^2 2p^2$ ☒ (c) $1s^2 2s^2 2p^4$ (d) $1s^2 2s^1 2p^3$ (e) $1s^2 2s^2$
18. Electrons have a set of quantum numbers that give them an identity or uniqueness in the atom. How many quantum numbers can they have?
(a) 1 ☒ (b) 2 (c) 3 (d) 4
19. Perform the operation: $3.456 + 12.1 + 0.0045$. Report with correct decimal places.
(a) 15.5605 ☒ (b) 15.56 (c) 15.6 (d) 15.561 (e) none of the above
20. The label on the bottle has the formula FeCl_3 . What is its name using the Stock system?
(a) iron chloride (b) iron(II) chloride ☒ (c) iron(III) chloride
(d) iron(III) chlorate (e) iron monochloride
21. One of the quantum numbers of an electron is related to the shape of the electron's orbital and is represented by a letter in this set: $\{s, p, d, f\}$. Which quantum number is this?
(a) n ☒ (b) l (c) m_l (d) m_s (e) Z
22. Which of these subatomic particles is **NOT** found in the nucleus of the atom?
(a) proton (b) neutron ☒ (c) electron
(d) all of choice (a)-(c) (e) none of choices (a)-(c)
23. What is a true statement regarding an element in the Periodic Table?
(a) the number of protons in its nucleus can be used to name the element
☒ (b) the atomic number (Z value) indicates the number of protons it has
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(d) all of the above are true
24. You have a volume of 650.0 mL. One and only one of the values below is equivalent and another way of expressing that value. Which one is it?
(a) $6.500 \times 10^3 \text{ L}$ ☒ (b) 0.650 L (c) 65 L (d) 6.5 L (e) 0.0006500 mL
25. What is the atomic number of zinc (Zn)?
(a) 65.39 (b) 1 (c) 29 ☒ (d) 30 (e) none of the above

26. A **cation** is an ion that has a net negative charge because it gained one or more electrons.

True ☒ False

27. Ionic compounds are typically formed between metals and nonmetals.

☒ True False

28. **Visible light** has a higher frequency and shorter wavelength than **microwaves**.

True ☒ False

29. The **p-block** of elements in the Periodic Table starts in Period 1.

True ☒ False

30. The **Pauli Exclusion Principle** states that an orbital can have only two electrons which are paired up with opposite spins

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31. According to the rules for significant figures, **conversion factors** (like $1\text{ m} = 100\text{ cm}$) are considered to have infinite significant digits and do not affect the precision of a calculation.

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(a) $^{12}_6\text{C}$ (b) $^{14}_6\text{C}$ (c) $^{14}_6\text{C}$ (d) $^{60}_{27}\text{Co}$
- Convert 4.56×10^{-3} into standard notation
(a) 0.00456 (b) 4560 (c) 0.000456 (d) 4,560
- Which of the following is the electron configuration for hydrogen?
(a) $1s^1$ (b) $1s^2$ (c) $1s^2 2s^1$ (d) $1p^1$ (e) $1s^2 2p^1$
- Which of the following is true about element bromine (Br)?
(a) It is a member of the p-block of elements
(b) It is a Group 17 element
(c) It is a non-metal
(d) all of the above
(e) none of the above
- What term describes an electron normally in $n = 1$ shell acquiring energy to be in $n = 2$ shell?
(a) emission (b) excited state (c) desorption (d) ground state (e) nucleated
- Express 0.0000721 in scientific notation with correct significant figures.
(a) 7.21×10^{-4} (b) 7.21×10^{-5} (c) 7.21×10^{-6} (d) 7.21×10^{-7} (e) none of the above
- When you complete the operation on the expression $1.450 \times 3.01 \div 10.025$, how many significant digits will the final result have?
(a) 2 (b) 3 (c) 4 (d) 5 (e) 7

15. The scientific method includes a stage where a **tentative explanation** is given to account for a set of related observations. What is that stage?
(a) law (b) hypothesis (c) theory (d) experimentation (e) the Atomic Model
16. Which of these is a polyatomic ion?
(a) Ne (b) H^+ (c) CO_3^{2-} (d) Br (e) Na^+
17. Which of these is the ground-state electron configuration for the **carbon (C)** atom?
(a) $1s^2 2s^2 2p^6$ (b) $1s^2 2s^2 2p^2$ (c) $1s^2 2s^2 2p^4$ (d) $1s^2 2s^1 2p^3$ (e) $1s^2 2s^2$
18. Electrons have a set of quantum numbers that give them an identity or uniqueness in the atom. How many quantum numbers can they have?
(a) 1 (b) 2 (c) 3 (d) 4
19. Perform the operation: $3.456 + 12.1 + 0.0045$. Report with correct decimal places.
(a) 15.5605 (b) 15.56 (c) 15.6 (d) 15.561 (e) none of the above
20. The label on the bottle has the formula FeCl_3 . What is its name using the Stock system? *Fe Cl^-*
(a) iron chloride (b) iron(II) chloride (c) iron(III) chloride
(d) iron(III) chlorate (e) iron monochloride
21. One of the quantum numbers of an electron is related to the shape of the electron's orbital and is represented by a letter in this set: $\{s, p, d, f\}$. Which quantum number is this?
(a) n (b) l (c) m_l (d) m_s (e) Z
22. Which of these subatomic particles is **NOT** found in the nucleus of the atom?
(a) proton (b) neutron (c) electron
(d) all of choice (a)-(c) (e) none of choices (a)-(c)
23. What is a true statement regarding an element in the Periodic Table?
(a) the number of protons in its nucleus can be used to name the element
(b) the atomic number (Z value) indicates the number of protons it has
(c) the mass number (A value) indicates the number of neutrons and protons it has
(d) all of the above are true
24. You have a volume of 650.0 mL. One and only one of the values below is equivalent and another way of expressing that value. Which one is it?
(a) $6.500 \times 10^3 \text{ L}$ (b) 0.650 L (c) 65 L (d) 6.5 L (e) 0.0006500 mL
25. What is the atomic number of zinc (Zn)?
(a) 65.39 (b) 1 (c) 29 (d) 30 (e) none of the above

26. A **cation** is an ion that has a net negative charge because it gained one or more electrons.

True False

27. Ionic compounds are typically formed between metals and nonmetals.

True False

28. **Visible light** has a higher frequency and shorter wavelength than **microwaves**.

True False

29. The **p-block** of elements in the Periodic Table starts in Period 1.

True False

30. The **Pauli Exclusion Principle** states that an orbital can have only two electrons which are paired up with opposite spins.

True False

31. According to the rules for significant figures, **conversion factors** (like $1\text{ m} = 100\text{ cm}$) are considered to have infinite significant digits and do not affect the precision of a calculation.

True False

32. The atomic number (Z) represents the number of protons in an atom.

True False

33. Electrons are found in the nucleus of the atom.

True False

34. Noble gases have complete valence shells and generally do not form ions.

True False

35. The law of conservation of mass states that matter **CAN** be created or destroyed in chemical reactions.

True False

Indicate (by circling or underlining carefully) *one* and *only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

- Which element belongs to the **alkali metals** group?
(a) calcium (b) sodium (c) magnesium (d) chlorine
- Which subatomic particle has a mass of approximately **1 amu** and a positive charge?
(a) electron (b) proton (c) neutron (d) none of the above
- Which of the following is a **polyatomic ion**?
(a) Cl^- (b) Na^+ (c) SO_4^{2-} (d) O (e) none of the above
- Convert 25.0°C into Kelvin.
(a) 248 K (b) 273 K (c) 298 K (d) 300 K
- A sample has a mass of 12.5 g and a volume of 2.50 mL. What is its density?
(a) 0.200 g/mL (b) 5.00 g/mL (c) 31.3 g/mL (d) 0.050 g/mL
- Which of the following is the correct name for $\text{Ca}(\text{NO}_3)_2$?
(a) cobalt fluoride (b) calcium nitroxide (c) calcium nitrate
(d) calcium oxide (e) kryptonite
- Which of the following terms describes the distance between two wave crests?
(a) wave speed (b) frequency (c) wavelength (d) amplitude
- Which of the following is the correct atomic notation for carbon-14?
(a) $^{12}_6\text{C}$ (b) $^{14}_6\text{C}$ (c) $^{14}_4\text{C}$ (d) $^{60}_{27}\text{Co}$
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 (a) $6.500 \times 10^3 \text{ L}$ (b) 0.650 L (c) 65 L (d) 6.5 L (e) 0.0006500 mL
25. What is the atomic number of zinc (Zn)?
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26. A **cation** is an ion that has a net negative charge because it gained one or more electrons.

True ☒ False

27. Ionic compounds are typically formed between metals and nonmetals.

☒ True False

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- Which of the following is a **polyatomic ion**?
(a) Cl^- (b) Na^+ (c) SO_4^{2-} (d) O (e) none of the above
- Convert 25.0°C into Kelvin. $25.0 + 273 = 298\text{K}$
(a) 248 K (b) 273 K (c) 298 K (d) 300 K
- A sample has a mass of 12.5 g and a volume of 2.50 mL. What is its density? $\frac{12.5\text{g}}{2.50\text{mL}}$
(a) 0.200 g/mL (b) 5.00 g/mL (c) 31.3 g/mL (d) 0.050 g/mL
- Which of the following is the correct name for $\text{Ca}(\text{NO}_3)_2$?
(a) cobalt fluoride (b) calcium nitroxide (c) calcium nitrate (d) calcium oxide (e) kryptonite
- Which of the following terms describes the distance between two wave crests?
(a) wave speed (b) frequency (c) wavelength (d) amplitude
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- Convert 4.56×10^{-3} into standard notation
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- Express 0.0000721 in scientific notation with correct significant figures. 7.21×10^{-5}
(a) 7.21×10^{-4} (b) 7.21×10^{-5} (c) 7.21×10^{-6} (d) 7.21×10^{-7} (e) none of the above
- When you complete the operation on the expression $1.450 \times \frac{3.01}{3} \div 10.025$, how many significant digits will the final result have?
(a) 2 (b) 3 (c) 4 (d) 5 (e) 7

15. The scientific method includes a stage where a **tentative explanation** is given to account for a set of related observations. What is that stage?
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25. What is the atomic number of zinc (Zn)?
 (a) 65.39 (b) 1 (c) 29 (d) 30 (e) none of the above

$$650.0 \text{ mL} \times \frac{1 \text{ L}}{1000 \text{ mL}}$$

26. A **cation** is an ion that has a net negative charge because it gained one or more electrons.

☒ True ☐ False

27. Ionic compounds are typically formed between metals and nonmetals.

☒ True ☐ False

28. **Visible light** has a higher frequency and shorter wavelength than **microwaves**.

True ☒ False

29. The **p-block** of elements in the Periodic Table starts in Period 1.

☒ True ☐ False

30. The **Pauli Exclusion Principle** states that an orbital can have only two electrons which are paired up with opposite spins

True ☒ False

31. According to the rules for significant figures, **conversion factors** (like $1\text{ m} = 100\text{ cm}$) are considered to have infinite significant digits and do not affect the precision of a calculation.

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32. The atomic number (Z) represents the number of protons in an atom.

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True ☒ False

34. Noble gases have complete valence shells and generally do not form ions

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35. The law of conservation of mass states that matter **CAN** be created or destroyed in chemical reactions.

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Indicate (by circling or underlining carefully) *one and only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

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(a) Cl^- (b) Na^+ (c) SO_4^{2-} (d) O (e) none of the above
- Convert 25.0°C into Kelvin.
(a) 248 K (b) 273 K (c) 298 K (d) 300 K
- A sample has a mass of 12.5 g and a volume of 2.50 mL. What is its density?
(a) 0.200 g/mL (b) 5.00 g/mL (c) 31.3 g/mL (d) 0.050 g/mL
 $D = \frac{m}{V}$
 $D = \frac{12.5}{2.50}$
- Which of the following is the correct name for $\text{Ca}(\text{NO}_3)_2$?
(a) cobalt fluoride (b) calcium nitroxide (c) calcium nitrate (d) calcium oxide (e) kryptonite
- Which of the following terms describes the distance between two wave crests?
(a) wave speed (b) frequency (c) wavelength (d) amplitude
- Which of the following is the correct atomic notation for carbon-14?
(a) $^{12}_6\text{C}$ (b) $^{14}_6\text{C}$ (c) $^{14}_{14}\text{C}$ (d) $^{60}_{27}\text{Co}$
- Convert 4.56×10^{-3} into standard notation
(a) 0.00456 (b) 4560 (c) 0.000456 (d) 4,560
- Which of the following is the electron configuration for hydrogen?
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(e) none of the above
- What term describes an electron normally in $n = 1$ shell acquiring energy to be in $n = 2$ shell?
(a) emission (b) excited state (c) desorption (d) ground state (e) nucleated
- Express 0.0000721 in scientific notation with correct significant figures.
(a) 7.21×10^{-4} (b) 7.21×10^{-5} (c) 7.21×10^{-6} (d) 7.21×10^{-7} (e) none of the above
- When you complete the operation on the expression $1.450 \times 3.01 \div 10.025$, how many significant digits will the final result have?
(a) 2 (b) 3 (c) 4 (d) 5 (e) 7

15. The scientific method includes a stage where a **tentative explanation** is given to account for a set of related observations. What is that stage?

- (a) law (b) hypothesis (c) theory (d) experimentation (e) the Atomic Model

16. Which of these is a polyatomic ion?

- (a) Ne (b) H^+ (c) CO_3^{2-} (d) Br (e) Na^+

17. Which of these is the ground-state electron configuration for the **carbon (C)** atom?

- ~~(a) $1s^2 2s^2 2p^5$~~ (b) $1s^2 2s^2 2p^2$ ~~(c) $1s^2 2s^2 2p^4$~~ (d) $1s^2 2s^1 2p^3$ ~~(e) $1s^2 2s^2$~~

18. Electrons have a set of quantum numbers that give them an identity or uniqueness in the atom. How many quantum numbers can they have?

- (a) 1 (b) 2 (c) 3 (d) 4

19. Perform the operation: $3.456 + 12.1 + 0.0045$. Report with correct decimal places.

- (a) 15.5605 (b) 15.56 (c) 15.6 (d) 15.561 (e) none of the above

20. The label on the bottle has the formula $FeCl_3$. What is its name using the Stock system?

- ~~(a) iron chloride~~ ~~(b) iron(II) chloride~~ (c) iron(III) chloride
(d) iron(III) chlorate ~~(e) iron monochloride~~

21. One of the quantum numbers of an electron is related to the shape of the electron's orbital and is represented by a letter in this set: {s, p, d, f}. Which quantum number is this?

- ~~(a) n~~ (b) l (c) m_l ~~(d) m_s~~ ~~(e) Z~~

22. Which of these subatomic particles is **NOT** found in the nucleus of the atom?

- (a) proton (b) neutron (c) electron
(d) all of choice (a)-(c) (e) none of choices (a)-(c)

23. What is a true statement regarding an element in the Periodic Table?

- (a) the number of protons in its nucleus can be used to name the element
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24. You have a volume of 650.0 mL. One and only one of the values below is equivalent and another way of expressing that value. Which one is it?

- ~~(a) 6.500×10^3 L~~ (b) 0.650 L (c) 65 L (d) 6.5 L ~~(e) 0.0006500 mL~~

25. What is the atomic number of zinc (Zn)?

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26. A **cation** is an ion that has a net negative charge because it gained one or more electrons.

True False

27. Ionic compounds are typically formed between metals and nonmetals.

True False

28. **Visible light** has a higher frequency and shorter wavelength than **microwaves**.

True False

29. The **p-block** of elements in the Periodic Table starts in Period 1.

True False

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True False

31. According to the rules for significant figures, **conversion factors** (like $1\text{ m} = 100\text{ cm}$) are considered to have infinite significant digits and do not affect the precision of a calculation.

True False

32. The atomic number (Z) represents the number of protons in an atom.

True False

33. Electrons are found in the nucleus of the atom.

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True False

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True False

Sections 43957-43958

Midterm Examination #1

Indicate (by circling or underlining carefully) *one* and *only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which element belongs to the **alkali metals** group?

- (a) calcium ☒ (b) sodium (c) magnesium (d) chlorine

2. Which subatomic particle has a mass of approximately **1 amu** and a positive charge?

- (a) electron ☒ (b) proton (c) neutron (d) none of the above

3. Which of the following is a **polyatomic ion**?

- (a) Cl^- (b) Na^+ ☒ (c) SO_4^{2-} (d) O (e) none of the above

4. Convert 25.0°C into Kelvin.

- (a) 248 K (b) 273 K ☒ (c) 298 K (d) 300 K

5. A sample has a mass of 12.5 g and a volume of 2.50 mL. What is its density? $\rightarrow$ pressure $d = \frac{\text{mass}}{\text{vol}}$

- (a) 0.200 g/mL ☒ (b) 5.00 g/mL (c) 31.3 g/mL (d) 0.050 g/mL

6. Which of the following is the correct name for $\text{Ca}(\text{NO}_3)_2$?

- (a) cobalt fluoride ☒ (b) calcium nitroxide (c) calcium nitrate
(d) calcium oxide (e) kryptonite

7. Which of the following terms describes the distance between two wave crests?

- (a) wave speed (b) frequency (c) wavelength ☒ (d) amplitude

8. Which of the following is the correct atomic notation for carbon-14?

- (a) $^{12}_6\text{C}$ (b) $^{14}_6\text{C}$ ☒ (c) $^{6}_{14}\text{C}$ (d) $^{60}_{27}\text{Co}$

9. Convert 4.56×10^{-3} into standard notation

- ☒ (a) 0.00456 (b) 4560 (c) 0.000456 (d) 4,560

10. Which of the following is the electron configuration for hydrogen?

- ☒ (a) $1s^1$ (b) $1s^2$ (c) $1s^2 2s^1$ (d) $1p^1$ (e) $1s^2 2p^1$

11. Which of the following is true about element bromine (Br)?

- (a) It is a member of the p-block of elements
(b) It is a Group 17 element
(c) It is a non-metal
☒ (d) all of the above
(e) none of the above

12. What term describes an electron normally in $n = 1$ shell acquiring energy to be in $n = 2$ shell?

- (a) emission (b) excited state (c) desorption ☒ (d) ground state (e) nucleated

13. Express 0.0000721 in scientific notation with correct significant figures.

- (a) 7.21×10^{-4} ☒ (b) 7.21×10^{-5} (c) 7.21×10^{-6} (d) 7.21×10^{-7} (e) none of the above

14. When you complete the operation on the expression $1.450 \times 3.01 \div 10.025$, how many significant digits will the final result have?

- (a) 2 ☒ (b) 3 (c) 4 (d) 5 (e) 7

15. The scientific method includes a stage where a **tentative explanation** is given to account for a set of related observations. What is that stage?
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17. Which of these is the ground-state electron configuration for the **carbon (C)** atom?
 (a) 1s²2s²2p⁶ ☒ (b) 1s²2s²2p² (c) 1s²2s²2p⁴ (d) 1s²2s¹2p³ (e) 1s²2s²
18. Electrons have a set of quantum numbers that give them an identity or uniqueness in the atom. How many quantum numbers can they have?
 (a) 1 ☒ (b) 2 (c) 3 (d) 4
19. Perform the operation: 3.456 + 12.1 + 0.0045. Report with correct decimal places.
 (a) 15.5605 (b) 15.56 ☒ (c) 15.6 (d) 15.561 (e) none of the above
20. The label on the bottle has the formula FeCl₃. What is its name using the Stock system?
 (a) iron chloride (b) iron(II) chloride ☒ (c) iron(III) chloride
 (d) iron(III) chlorate (e) iron monochloride
21. One of the quantum numbers of an electron is related to the shape of the electron's orbital and is represented by a letter in this set: {s, p, d, f}. Which quantum number is this?
 (a) n (b) l (c) m_l ☒ (d) m_s (e) Z
22. Which of these subatomic particles is **NOT** found in the nucleus of the atom?
 (a) proton (b) neutron (c) electron
☒ (d) all of choice (a)-(c) (e) none of choices (a)-(c)
23. What is a true statement regarding an element in the Periodic Table?
 (a) the number of protons in its nucleus can be used to name the element
 (b) the atomic number (Z value) indicates the number of protons it has
 (c) the mass number (A value) indicates the number of neutrons and protons it has
☒ (d) all of the above are true
24. You have a volume of 6.500 mL. One and only one of the values below is equivalent and another way of expressing that value. Which one is it?
 (a) 6.500 x 10³ L (b) 0.650 L (c) 65 L ☒ (d) 6.5 L (e) 0.0006500 mL
25. What is the atomic number of zinc (Zn)?
 (a) 65.39 (b) 1 (c) 29 ☒ (d) 30 (e) none of the above

26. A **cation** is an ion that has a net negative charge because it gained one or more electrons.

True ☒ False

27. Ionic compounds are typically formed between metals and nonmetals.

☒ True False

28. **Visible light** has a higher frequency and shorter wavelength than **microwaves**.

☒ True False

29. The **p-block** of elements in the Periodic Table starts in Period 1.

True ☒ False

30. The **Pauli Exclusion Principle** states that an orbital can have only two electrons which are paired up with opposite spins.

☒ True False

31. According to the rules for significant figures, **conversion factors** (like $1\text{ m} = 100\text{ cm}$) are considered to have infinite significant digits and do not affect the precision of a calculation.

True ☒ False

32. The atomic number (Z) represents the number of protons in an atom.

☒ True False

33. Electrons are found in the nucleus of the atom.

True ☒ False

34. Noble gases have complete valence shells and generally do not form ions.

☒ True False

35. The law of conservation of mass states that matter **CAN** be created or destroyed in chemical reactions.

True ☒ False

Indicate (by circling or underlining carefully) *one and only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

- Which element belongs to the **alkali metals** group?
(a) calcium (b) ☒ sodium (c) magnesium (d) chlorine
- Which subatomic particle has a mass of approximately **1 amu** and a positive charge?
(a) electron (b) ☒ proton (c) neutron (d) none of the above
- Which of the following is a **polyatomic ion**?
(a) Cl^- (b) Na^+ (c) ☒ SO_4^{2-} (d) O (e) none of the above
- Convert 25.0°C into Kelvin.
(a) 248 K (b) 273 K (c) ☒ 298 K (d) 300 K
- A sample has a mass of 12.5 g and a volume of 2.50 mL. What is its density?
(a) 0.200 g/mL (b) ☒ 5.00 g/mL (c) 31.3 g/mL (d) 0.050 g/mL
- Which of the following is the correct name for $\text{Ca}(\text{NO}_3)_2$?
(a) cobalt fluoride (b) calcium nitroxide (c) ☒ calcium nitrate
(d) calcium oxide (e) kryptonite
- Which of the following terms describes the distance between two wave crests?
(a) wave speed (b) frequency (c) wavelength (d) ☒ amplitude
- Which of the following is the correct atomic notation for carbon-14?
(a) $^{12}_6\text{C}$ (b) ☒ $^{14}_6\text{C}$ (c) $^{14}_7\text{C}$ (d) $^{60}_{27}\text{Co}$
- Convert 4.56×10^{-3} into standard notation
(a) ☒ 0.00456 (b) 4560 (c) 0.000456 (d) 4,560
- Which of the following is the electron configuration for hydrogen?
(a) ☒ $1s^1$ (b) $1s^2$ (c) $1s^2 2s^1$ (d) $1p^1$ (e) $1s^2 2p^1$
- Which of the following is true about element bromine (Br)?
(a) It is a member of the p-block of elements
(b) It is a Group 17 element
(c) It is a non-metal
(d) ☒ all of the above
(e) none of the above
- What term describes an electron normally in $n = 1$ shell acquiring energy to be in $n = 2$ shell?
(a) emission (b) ☒ excited state (c) desorption (d) ground state (e) nucleated
- Express 0.0000721 in scientific notation with correct significant figures.
(a) 7.21×10^{-4} (b) ☒ 7.21×10^{-5} (c) 7.21×10^{-6} (d) 7.21×10^{-7} (e) none of the above
- When you complete the operation on the expression $1.450 \times 3.01 \div 10.025$, how many significant digits will the final result have?
(a) 2 (b) ☒ 3 (c) 4 (d) 5 (e) 7

S 0
P 1 3
D 2 5
F 3 7

15. The scientific method includes a stage where a **tentative explanation** is given to account for a set of related observations. What is that stage?
 (a) law ☒ (b) hypothesis (c) theory (d) experimentation (e) the Atomic Model
16. Which of these is a polyatomic ion?
 (a) Ne (b) H^+ ☒ (c) CO_3^{2-} (d) Br (e) Na^+
17. Which of these is the ground-state electron configuration for the **carbon (C)** atom?
 (a) $1s^2 2s^2 2p^5$ ☒ (b) $1s^2 2s^2 2p^2$ (c) $1s^2 2s^2 2p^4$ (d) $1s^2 2s^1 2p^3$ (e) $1s^2 2s^2$
18. Electrons have a set of quantum numbers that give them an identity or uniqueness in the atom. How many quantum numbers can they have?
 (a) 1 (b) 2 (c) 3 ☒ (d) 4
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 (b) the atomic number (Z value) indicates the number of protons it has
☒ (c) the mass number (A value) indicates the number of neutrons and protons it has
 (d) all of the above are true
24. You have a volume of 650.0 mL. One and only one of the values below is equivalent and another way of expressing that value. Which one is it?
 (a) $6.500 \times 10^3 \text{ L}$ ☒ (b) 0.650 L (c) 65 L (d) 6.5 L (e) 0.0006500 mL
25. What is the atomic number of zinc (Zn)?
 (a) 65.39 (b) 1 (c) 29 ☒ (d) 30 (e) none of the above

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True ☒ False

Chemistry 3A – Fall 2025

Sections 43957-43958

Midterm Examination #3

Rogelio Macanilla Portner

0977963

RETAKED VERSION

Indicate (by circling or underlining carefully) *one and only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

- Which of the gas laws states the volume of a gas is directly proportional to amount of a gas in moles ($V=k \times n$)?
(a) Boyle's Law (b) Charles' Law (c) Gay-Lussac's Law
(d) Avogadro's Law (e) Ideal Gas Law
- In a redox reaction, the species that **loses electrons** is said to be:
(a) reduced (b) precipitated (c) oxidized (d) dissolved (e) unchanged
- A chemical reaction where two or more simple reactants combine to form a single, more complex product (e.g., $A + B \rightarrow AB$) is what type of reaction?
(a) decomposition (b) single displacement (SD) (c) double displacement (DD)
(d) combination (synthesis) reaction (e) combustion reaction
- What is molarity of aqueous solution in which 0.50 moles KCl dissolved to final 500.0 mL volume?
(a) 0.25 M (b) 0.50 M (c) 1.0 M (d) 2.0 M (e) 0.0010 M
- If you take 10.0 mL of a 2.5 M stock solution of NaCl and dilute it with water to a final volume of 50.0 mL, what is the final concentration (C_2)?
(a) 0.10 M (b) 0.50 M (c) 1.25 M (d) 5.0 M (e) 0.050 M
- A solution is prepared by mixing 10.0 g of sugar (solute) with 90.0 g of water (solvent). What is the percent by mass of the sugar in the solution?
(a) 1.0% (b) 10.0% (c) 11.1% (d) 90.0% (e) 100.0%
- Calculate the **pressure** of $n = 2.00$ mol of gas at $T = 273$ K, $V = 44.8$ L. ($R = 0.0821$ L·atm/[mol·K])
(a) 1.00 atm (b) 2.00 atm (c) 0.500 atm (d) 22.4 atm (e) none of the above
- Convert a **pressure** of 760 mmHg to atm.
(a) 0.500 atm (b) 1.00 atm (c) 7.60 atm (d) 76.0 atm
- Which law states that **pressure** and **volume** are inversely related, that is $P_1V_1 = P_2V_2$?
(a) Boyle's Law (b) Charles' Law (c) Gay-Lussac's Law (d) Avogadro's Law
- Which of the following is a **combustion** reaction?
(a) $\text{CH}_4 + 2 \text{O}_2 \rightarrow \text{CO}_2 + 2 \text{H}_2\text{O}$ (b) $\text{NaCl} + \text{AgNO}_3 \rightarrow \text{AgCl} + \text{NaNO}_3$
(c) $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$ (d) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
(e) none of the above
- Which of the following is the correct oxidation number of **hydrogen** in H_2O molecule?
(a) 0 (b) -1 (c) +1 (d) +2 (e) none of the above
- A substance that completely dissociates into ions when dissolved in an aqueous solution and conducts electricity strongly is classified as a:
(a) non-electrolyte (b) weak electrolyte (c) strong electrolyte
(d) saturated solution (e) suspension

13. What is the oxidation number (ON) for a neutral atom of Cl in its elemental state, Cl_2 ?
 (a) -1 (b) 0 (c) +1 (d) +2 (e) +7
14. The strongest type of intermolecular force that exists between two molecules of H_2O (water) is:
 (a) dispersion forces (London) (b) dipole-dipole forces (c) ion-dipole forces
 (d) hydrogen bonding (e) covalent bonding
15. How many moles are in 15.0 g of KOH? (molar mass of KOH = 56.11 g/mol)
 (a) 0.188 mol (b) 0.267 mol (c) 0.459 mol (d) 1.25 mol
16. Which is a key assumption of the **Kinetic Molecular Theory** for an *ideal gas*?
 (a) gas molecules exert significant attractive forces on each other
 (b) the molecules that make up the gas phase must interact with their liquid phase
 (c) gas collisions are completely elastic (energy is not lost in collision)
 (d) the pressure of the gas is inversely proportional to temperature
17. Which of the following salts is generally considered **soluble** in water?
 (a) AgCl (b) PbI_2 (c) BaSO_4 (d) KNO_3 (e) CaCO_3
18. A solution that contains the maximum amount of dissolved solute possible at a given temperature is called a:
 (a) unsaturated solution (b) saturated solution (c) supersaturated solution
 (d) dilute solution (e) concentrated solution
19. A gas has $P_1 = 2.00$ atm, $V_1 = 4.00$ L. If pressure increases to $P_2 = 8.00$ atm, what is V_2 ?
 (a) 0.50 L (b) 1.00 L (c) 2.00 L (d) 16.0 L (e) none of the above
20. Hydrogen bonding can only occur between molecules where hydrogen is directly bonded to one of three highly electronegative atoms. These atoms are:
 (a) C, H, O (b) S, Cl, Br (c) Cl, Br, I (d) N, O, F (e) Na, K, Li
21. In a sugar-water solution, the sugar is the **solute** and the water is the **solvent**. The **solvent** is defined as the component present in:
 (a) the smallest amount (b) the solid phase (c) the greatest amount
 (d) the gaseous phase (e) the ionic form
22. The concentration unit of **molarity (M)** is correctly expressed in which of the following units?
 (a) mol/kg (b) g/L (c) mol/mL (d) mol/L (e) kg/mol
23. Which of the following is a spectator ion in the reaction:
 $\text{NaCl (aq)} + \text{AgNO}_3 \text{ (aq)} \rightarrow \text{AgCl (s)} + \text{NaNO}_3 \text{ (aq)}$?
 (a) Na^+ (b) Cl^- (c) Ag^+ (d) NO_3^- (e) both (a) and (d)
24. Which statement best describes "like dissolves like"?
 (a) polar solutes dissolve in polar solvents (b) nonpolar solutes dissolve in nonpolar solvents
 (c) ionic solutes dissolve in polar solvents (d) all of the above
 (e) none of the above
25. A precipitate forms and you have to write the chemical reaction equation. How will you indicate the compound that precipitated?
 (a) (s) (b) (l) (c) (g) (d) (aq) (e) none of these

Circle either **True** or **False** for the following statements:

26. A solution that conducts electricity must contain ions.

☒ True ☐ False

27. Combustion reactions always involve oxygen as a product.

☒ True ☐ False

28. In a balanced chemical equation, the number and type of atoms must be equal on both sides.

☒ True ☐ False

29. The oxidation number of a pure element is always zero.

☒ True ☐ False

30. In a chemical reaction equation, compounds to the right of the reaction arrow are called

reactants.

True ☒ False

31. The oxidation number of any Group 1 metal (like Na or K) in any compound is always +1.

☒ True ☐ False

32. The total pressure of a gas mixture is equal to the sum of the partial pressures of all the individual gases in the mixture.

☒ True ☐ False

33. The solid that forms from a solution during a chemical reaction, often by mixing two solutions, is called a **precipitate**.

True ☒ False

34. At STP, the temperature is 25 °C and pressure is 1 atm.

☒ True ☐ False

35. The variable R in the Ideal Gas Law equation ($PV=nRT$) is known as the **universal gas constant**.

☒ True ☐ False

Indicate (by circling or underlining carefully) *one and only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. How many **valence electrons** does a neutral atom of **nitrogen** (N) have? (nitrogen is Group 15)
(a) 2 (b) 3 ☒ (c) 5 (d) 7
2. Which crystalline solid is ice (H_2O solid)?
(a) ionic (b) metallic (c) covalent network ☒ (d) molecular (e) none of the above
3. Which term describes the upward movement of liquid in a narrow tube?
☒ (a) adhesion (b) cohesion (c) capillary action (d) viscosity
4. Which process is endothermic?
(a) freezing water (b) condensation of steam (c) melting ice
(d) deposition of CO_2 ☒ (e) none of the above
5. Which molecular geometry describes NH_3 ?
(a) linear (b) trigonal planar (c) trigonal pyramidal ☒ (d) tetrahedral
6. Avogadro's number represents:
(a) number of grams in a mole ☒ (b) number of atoms/molecules in a mole
(c) number of electrons in an atom (d) number of protons in a nucleus
7. Lewis structures represent:
(a) protons in the nucleus (b) neutrons in the nucleus ☒ (c) valence electrons of atoms
(d) atomic mass units (e) none of the above
8. The **octet rule** states that atoms want how many electrons in their valence shell?
(a) 2 (b) 6 ☒ (c) 8 (d) 10 (e) none of the above
9. Which intermolecular force is strongest?
(a) London dispersion (b) dipole-dipole ☒ (c) hydrogen bonding (d) none of the above
10. Which crystalline solid is copper metal?
(a) ionic ☒ (b) metallic (c) covalent network (d) molecular
11. Which molecule can form hydrogen bonds?
☒ (a) CH_4 (b) CO_2 (c) NH_3 (d) CCl_4
12. Which crystalline solid is NaCl ?
(a) ionic (b) metallic (c) covalent network (d) molecular ☒ (e) none of the above
13. The Valence Shell Electron Pair Repulsion (VSEPR) theory is used to predict:
☒ (a) the mass of the electron (b) the molar mass of a molecule
(c) the three-dimensional shape of a molecule (d) The percent composition of a compound.
(e) The atomic number of an element.

14. Which of the following represents the enthalpy (energy) change required to convert a solid substance directly into a gas?
- (a) ΔH_{vap} (b) ΔH_{fus} (c) ΔH_{sub} (d) ΔH_{melt} (e) ΔH_{comb}
15. The term that defines the mass of one mole of a substance and has units of g/mol is:
- (a) atomic number (b) molar mass (c) specific heat
(d) Avogadro's number (e) empirical formula mass
16. Which phase of matter is characterized by having a **definite volume** but an **indefinite shape**?
- (a) gas (b) liquid (c) solid (d) plasma (e) fluid
17. Which of the following compounds is most likely to be formed through **ionic bonding**?
- (a) CO_2 (b) NH_3 (c) PCl_3 (d) CaO (e) CH_4
18. Convert 100.0 cal to joules. (1 cal = 4.184 J)
- (a) 418 J (b) 41.8 J (c) 4.18 J (d) 0.418 J (e) none of the above
19. A compound has 40.0 g C, 6.7 g H, and 53.3 g O. What is its empirical formula?
- (a) CHO (b) CH_2O (c) $\text{C}_2\text{H}_4\text{O}_2$ (d) $\text{C}_3\text{H}_6\text{O}_3$
20. Calculate the molar mass of Na_2CO_3 .
- (a) 106 g/mol (b) 62 g/mol (c) 44 g/mol (d) 23 g/mol (e) none of the above
21. How many atoms are in 2.00 mol of carbon?
- (a) 2.00×10^{23} (b) 3.01×10^{23} (c) 6.022×10^{23} (d) 1.20×10^{24}
22. A 50.0 g sample of aluminum is heated from 25.0 °C to 75.0 °C. Aluminum has a heat capacity $c_p = 0.900 \text{ J/g} \cdot ^\circ\text{C}$. How much heat is absorbed?
- (a) 2250 J (b) 4500 J (c) 900 J (d) 75 J
23. The molar mass of copper sulfate pentahydrate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) is 250 g/mol. What percent of the mass is water (H_2O)?
- (a) 18% (b) 25% (c) 36% (d) 45% (e) none of the above
24. As you move down a group (from **top to bottom** of periodic table), the atomic size (radius) of the elements generally:
- (a) decreases (b) increases (c) stays the same (d) becomes ionic (e) doubles
25. London Dispersion Forces (LDF) are the only type of intermolecular force present in molecules that are:
- (a) ionic (b) highly polar (c) capable of hydrogen bonding
(d) nonpolar (e) solids at room temperature

$$100.0 \text{ cal} \times \frac{4.184 \text{ J}}{1 \text{ cal}} = 418.4 \text{ J}$$

$$159.41 + 90.8 = 36.2$$

$$290.41$$

Circle either **True** or **False** for the following statements:

26. The burning of gasoline is an exothermic process.

☒ True ☐ False

27. Metals typically achieve stability by sharing their valence electrons in a "sea of electrons," resulting in metallic bonding.

True ☒ False

28. The molar mass of an element is numerically equal to its average atomic mass found on the periodic table.

True ☒ False

29. During a phase change (like melting or boiling), adding heat to a substance will cause its temperature to momentarily stop increasing.

True ☒ False

30. A polar bond occurs when two atoms have identical electronegativity.

True ☒ False

31. London dispersion forces occur only in polar molecules.

☒ True ☐ False

32. VSEPR theory is used to predict molecular geometry.

True ☒ False

33. The first ionization energy generally increases as you move down a group (column) in the Periodic Table.

☒ True ☐ False

34. Kinetic energy is the energy an object possesses due to its motion or velocity.

☒ True ☐ False

35. The percent by mass of all elements in a compound must always sum up to 100%

☒ True ☐ False

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22. A 50.0 g sample of aluminum is heated from 25.0 °C to 75.0 °C. Aluminum has a heat capacity $c_p = 0.900 \text{ J/g}\cdot^\circ\text{C}$. How much heat is absorbed?

- (a) 2250 J (b) 4500 J (c) 900 J (d) 75 J

23. The molar mass of copper sulfate pentahydrate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) is 250 g/mol. What percent of the mass is water (H_2O)?

- (a) 18% (b) 25% (c) 36% (d) 45% (e) none of the above

24. As you move down a group (from **top to bottom** of periodic table), the atomic size (radius) of the elements generally:

- (a) decreases (b) increases (c) stays the same (d) becomes ionic (e) doubles

25. London Dispersion Forces (LDF) are the only type of intermolecular force present in molecules that are:

- (a) ionic (b) highly polar (c) capable of hydrogen bonding
(d) nonpolar (e) solids at room temperature

Circle either **True** or **False** for the following statements:

26. The burning of gasoline is an exothermic process.

☒ True ☐ False

27. Metals typically achieve stability by sharing their valence electrons in a "sea of electrons," resulting in metallic bonding.

True ☒ False

28. The molar mass of an element is numerically equal to its average atomic mass found on the periodic table.

☒ True ☐ False

29. During a phase change (like melting or boiling), adding heat to a substance will cause its temperature to momentarily stop increasing.

☒ True ☐ False

30. A polar bond occurs when two atoms have identical electronegativity.

True ☒ False

31. London dispersion forces occur only in polar molecules.

True ☒ False

32. VSEPR theory is used to predict molecular geometry.

☒ True ☐ False

33. The first ionization energy generally increases as you move down a group (column) in the Periodic Table.

☒ True ☐ False

34. Kinetic energy is the energy an object possesses due to its motion or velocity.

☒ True ☐ False

35. The percent by mass of all elements in a compound must always sum up to 100%

True ☒ False

Indicate (by circling or underlining carefully) *one* and *only one* choice that is **BEST** response for the question.
Point values: multiple choice 3 pt; true/false 1 pt

- Which of the gas laws states the volume of a gas is directly proportional to amount of a gas in moles ($V = k \times n$)?
(a) Boyle's Law (b) Charles' Law (c) Gay-Lussac's Law
(d) Avogadro's Law (e) Ideal Gas Law
- In a redox reaction, the species that **loses electrons** is said to be:
(a) reduced (b) precipitated (c) oxidized (d) dissolved (e) unchanged
- A chemical reaction where two or more simple reactants combine to form a single, more complex product (e.g., $A + B \rightarrow AB$) is what type of reaction?
(a) decomposition (b) single displacement (SD) (c) double displacement (DD)
(d) combination (synthesis) reaction (e) combustion reaction
- What is molarity of aqueous solution in which 0.50 moles KCl dissolved to final 500.0 mL volume?
(a) 0.25 M (b) 0.50 M (c) 1.0 M (d) 2.0 M (e) 0.0010 M
- If you take 10.0 mL of a 2.5 M stock solution of NaCl and dilute it with water to a final volume of 50.0 mL, what is the final concentration (C_2)?
(a) 0.10 M (b) 0.50 M (c) 1.25 M (d) 5.0 M (e) 0.050 M
- A solution is prepared by mixing 10.0 g of sugar (solute) with 90.0 g of water (solvent). What is the percent by mass of the sugar in the solution?
(a) 1.0% (b) 10.0% (c) 11.1% (d) 90.0% (e) 100.0%
- Calculate the **pressure** of $n = 2.00$ mol of gas at $T = 273$ K, $V = 44.8$ L. ($R = 0.0821$ L·atm/[mol·K])
(a) 1.00 atm (b) 2.00 atm (c) 0.500 atm (d) 22.4 atm (e) none of the above
- Convert a **pressure** of 760 mmHg to atm.
(a) 0.500 atm (b) 1.00 atm (c) 7.60 atm (d) 76.0 atm
- Which law states that **pressure** and **volume** are inversely related, that is $P_1V_1 = P_2V_2$?
(a) Boyle's Law (b) Charles' Law (c) Gay-Lussac's Law (d) Avogadro's Law
- Which of the following is a **combustion** reaction?
(a) $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$ (b) $\text{NaCl} + \text{AgNO}_3 \rightarrow \text{AgCl} + \text{NaNO}_3$
(c) $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$ (d) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
(e) none of the above
- Which of the following is the correct oxidation number of **hydrogen** in H_2O molecule?
(a) 0 (b) -1 (c) +1 (d) +2 (e) none of the above
- A substance that completely dissociates into ions when dissolved in an aqueous solution and conducts electricity strongly is classified as a:
(a) non-electrolyte (b) weak electrolyte (c) strong electrolyte
(d) saturated solution (e) suspension

13. What is the oxidation number (ON) for a neutral atom of Cl in its elemental state, Cl_2 ?
 (a) -1 (b) 0 (c) +1 (d) +2 (e) +7
14. The strongest type of intermolecular force that exists between two molecules of H_2O (water) is:
 (a) dispersion forces (London) (b) dipole-dipole forces (c) ion-dipole forces
 (d) hydrogen bonding (e) covalent bonding
15. How many moles are in 15.0 g of KOH? (molar mass of KOH = 56.11 g/mol)
 (a) 0.188 mol (b) 0.267 mol (c) 0.459 mol (d) 1.25 mol
16. Which is a key assumption of the **Kinetic Molecular Theory** for an *ideal gas*?
 (a) gas molecules exert significant attractive forces on each other
 (b) the molecules that make up the gas phase must interact with their liquid phase
 (c) gas collisions are completely elastic (energy is not lost in collision)
 (d) the pressure of the gas is inversely proportional to temperature
17. Which of the following salts is generally considered **soluble** in water?
 (a) AgCl (b) PbI_2 (c) BaSO_4 (d) KNO_3 (e) CaCO_3
18. A solution that contains the maximum amount of dissolved solute possible at a given temperature is called a:
 (a) unsaturated solution (b) saturated solution (c) supersaturated solution
 (d) dilute solution (e) concentrated solution
19. A gas has $P_1 = 2.00 \text{ atm}$, $V_1 = 4.00 \text{ L}$. If pressure increases to $P_2 = 8.00 \text{ atm}$, what is V_2 ?
 (a) 0.50 L (b) 1.00 L (c) 2.00 L (d) 16.0 L (e) none of the above
20. Hydrogen bonding can only occur between molecules where hydrogen is directly bonded to one of three highly electronegative atoms. These atoms are:
 (a) C, H, O (b) S, Cl, Br (c) Cl, Br, I (d) N, O, F (e) Na, K, Li
21. In a sugar-water solution, the sugar is the **solute** and the water is the **solvent**. The **solvent** is defined as the component present in:
 (a) the smallest amount (b) the solid phase (c) the greatest amount
 (d) the gaseous phase (e) the ionic form
22. The concentration unit of **molarity (M)** is correctly expressed in which of the following units?
 (a) mol/kg (b) g/L (c) mol/mL (d) mol/L (e) kg/mol
23. Which of the following is a spectator ion in the reaction:
 $\text{NaCl (aq)} + \text{AgNO}_3 \text{ (aq)} \rightarrow \text{AgCl (s)} + \text{NaNO}_3 \text{ (aq)}$?
 (a) Na^+ (b) Cl^- (c) Ag^+ (d) NO_3^- (e) both (a) and (d)
24. Which statement best describes "like dissolves like"?
 (a) polar solutes dissolve in polar solvents (b) nonpolar solutes dissolve in nonpolar solvents
 (c) ionic solutes dissolve in polar solvents (d) all of the above
 (e) none of the above
25. A precipitate forms and you have to write the chemical reaction equation. How will you indicate the compound that precipitated?
 (a) (s) (b) (l) (c) (g) (d) (aq) (e) none of these

Circle either **True** or **False** for the following statements:

26. A solution that conducts electricity must contain ions.

☒ True ☐ False

27. Combustion reactions always involve oxygen as a product.

☒ True ☐ False

28. In a balanced chemical equation, the number and type of atoms must be equal on both sides.

☒ True ☐ False

29. The oxidation number of a pure element is always zero.

☒ True ☐ False

30. In a chemical reaction equation, compounds to the right of the reaction arrow are called **reactants**.

True ☒ False

31. The oxidation number of any Group 1 metal (like Na or K) in any compound is always +1.

☒ True ☐ False

32. The total pressure of a gas mixture is equal to the sum of the partial pressures of all the individual gases in the mixture.

☒ True ☐ False

33. The solid that forms from a solution during a chemical reaction, often by mixing two solutions, is called a **precipitate**.

☒ True ☐ False

34. At STP, the temperature is 25 °C and pressure is 1 atm.

True ☒ False

35. The variable R in the Ideal Gas Law equation ($PV=nRT$) is known as the **universal gas constant**.

True ☒ False

Midterm Examination #3

Indicate (by circling or underlining carefully) *one* and *only one* choice that is **BEST** response for the question.
Point values: multiple choice 3 pt; true/false 1 pt

- Which of the gas laws states the volume of a gas is directly proportional to amount of a gas in moles ($V=k \times n$)?
(a) Boyle's Law (b) Charles' Law (c) Gay-Lussac's Law
(d) Avogadro's Law (e) Ideal Gas Law
- In a redox reaction, the species that **loses electrons** is said to be:
(a) reduced (b) precipitated (c) oxidized (d) dissolved (e) unchanged
- A chemical reaction where two or more simple reactants combine to form a single, more complex product (e.g., $A + B \rightarrow AB$) is what type of reaction?
(a) decomposition (b) single displacement (SD) (c) double displacement (DD)
(d) combination (synthesis) reaction (e) combustion reaction
- What is molarity of aqueous solution in which 0.50 moles KCl dissolved to final 500.0 mL volume?
(a) 0.25 M (b) 0.50 M (c) 1.0 M (d) 2.0 M (e) 0.0010 M
- If you take 10.0 mL of a 2.5 M stock solution of NaCl and dilute it with water to a final volume of 50.0 mL, what is the final concentration (C_2)?
(a) 0.10 M (b) 0.50 M (c) 1.25 M (d) 5.0 M (e) 0.050 M
- A solution is prepared by mixing 10.0 g of sugar (solute) with 90.0 g of water (solvent). What is the percent by mass of the sugar in the solution?
(a) 1.0% (b) 10.0% (c) 11.1% (d) 90.0% (e) 100.0%
- Calculate the **pressure** of $n = 2.00$ mol of gas at $T = 273$ K, $V = 44.8$ L. ($R = 0.0821$ L·atm/(mol·K))
(a) 1.00 atm (b) 2.00 atm (c) 0.500 atm (d) 22.4 atm (e) none of the above
- Convert a **pressure** of 760 mmHg to atm.
(a) 0.500 atm (b) 1.00 atm (c) 7.60 atm (d) 76.0 atm
- Which law states that **pressure** and **volume** are inversely related, that is $P_1V_1 = P_2V_2$?
(a) Boyle's Law (b) Charles' Law (c) Gay-Lussac's Law (d) Avogadro's Law
- Which of the following is a **combustion** reaction?
(a) $CH_4 + 2 O_2 \rightarrow CO_2 + 2 H_2O$ (b) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
(c) $HCl + NaOH \rightarrow NaCl + H_2O$ (d) $CaCO_3 \rightarrow CaO + CO_2$
(e) none of the above
- Which of the following is the correct oxidation number of **hydrogen** in H_2O molecule?
(a) 0 (b) -1 (c) +1 (d) +2 (e) none of the above
- A substance that completely dissociates into ions when dissolved in an aqueous solution and conducts electricity strongly is classified as a:
(a) non-electrolyte (b) weak electrolyte (c) strong electrolyte
(d) saturated solution (e) suspension

13. What is the oxidation number (ON) for a neutral atom of Cl in its elemental state, Cl_2 ?
 (a) -1 (b) 0 (c) +1 (d) +2 (e) +7
14. The strongest type of intermolecular force that exists between two molecules of H_2O (water) is:
 (a) dispersion forces (London) (b) dipole-dipole forces (c) ion-dipole forces
 (d) hydrogen bonding (e) covalent bonding
15. How many moles are in 15.0 g of KOH? (molar mass of KOH = 56.11 g/mol)
 (a) 0.188 mol (b) 0.267 mol (c) 0.459 mol (d) 1.25 mol
16. Which is a key assumption of the **Kinetic Molecular Theory** for an *ideal gas*?
 (a) gas molecules exert significant attractive forces on each other
 (b) the molecules that make up the gas phase must interact with their liquid phase
 (c) gas collisions are completely elastic (energy is not lost in collision)
 (d) the pressure of the gas is inversely proportional to temperature
17. Which of the following salts is generally considered **soluble** in water?
 (a) AgCl (b) PbI_2 (c) BaSO_4 (d) KNO_3 (e) CaCO_3
18. A solution that contains the maximum amount of dissolved solute possible at a given temperature is called a:
 (a) unsaturated solution (b) saturated solution (c) supersaturated solution
 (d) dilute solution (e) concentrated solution
19. A gas has $P_1 = 2.00 \text{ atm}$, $V_1 = 4.00 \text{ L}$. If pressure increases to $P_2 = 8.00 \text{ atm}$, what is V_2 ?
 (a) 0.50 L (b) 1.00 L (c) 2.00 L (d) 16.0 L (e) none of the above
20. Hydrogen bonding can only occur between molecules where hydrogen is directly bonded to one of three highly electronegative atoms. These atoms are:
 (a) C, H, O (b) S, Cl, Br (c) Cl, Br, I (d) N, O, F (e) Na, K, Li
21. In a sugar-water solution, the sugar is the **solute** and the water is the **solvent**. The solvent is defined as the component present in:
 (a) the smallest amount (b) the solid phase (c) the greatest amount
 (d) the gaseous phase (e) the ionic form
22. The concentration unit of **molarity (M)** is correctly expressed in which of the following units?
 (a) mol/kg (b) g/L (c) mol/mL (d) mol/L (e) kg/mol
23. Which of the following is a spectator ion in the reaction:
 $\text{NaCl (aq)} + \text{AgNO}_3 \text{ (aq)} \rightarrow \text{AgCl (s)} + \text{NaNO}_3 \text{ (aq)}$
 (a) Na^+ (b) Cl^- (c) Ag^+ (d) NO_3^- (e) both (a) and (d)
24. Which statement best describes "like dissolves like"?
 (a) polar solutes dissolve in polar solvents (b) nonpolar solutes dissolve in nonpolar solvents
 (c) ionic solutes dissolve in polar solvents (d) all of the above
 (e) none of the above
25. A precipitate forms and you have to write the chemical reaction equation. How will you indicate the compound that precipitated?
 (a) (s) (b) (l) (c) (g) (d) (aq) (e) none of these

Circle either **True** or **False** for the following statements:

26. A solution that conducts electricity must contain ions.

☒ True ☐ False

27. Combustion reactions always involve oxygen as a product.

True ☒ False

28. In a balanced chemical equation, the number and type of atoms must be equal on both sides.

☒ True ☐ False

29. The oxidation number of a pure element is always zero.

☒ True ☐ False

30. In a chemical reaction equation, compounds to the right of the reaction arrow are called **reactants**.

True ☒ False

31. The oxidation number of any Group 1 metal (like Na or K) in any compound is always +1.

True ☒ False

32. The total pressure of a gas mixture is equal to the sum of the partial pressures of all the individual gases in the mixture.

☒ True ☐ False

33. The solid that forms from a solution during a chemical reaction, often by mixing two solutions, is called a **precipitate**.

True ☒ False

34. At STP, the temperature is 25 °C and pressure is 1 atm.

☒ True ☐ False

35. The variable R in the Ideal Gas Law equation ($PV=nRT$) is known as the **universal gas constant**.

True ☒ False

Indicate (by circling or underlining carefully) *one and only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which of the gas laws states the volume of a gas is directly proportional to amount of a gas in moles ($V=k \times n$)?
(a) Boyle's Law (b) Charles' Law (c) Gay-Lussac's Law
(d) Avogadro's Law (e) Ideal Gas Law
2. In a redox reaction, the species that **loses electrons** is said to be:
(a) reduced (b) precipitated (c) oxidized (d) dissolved (e) unchanged "Ox. Rtg."
3. A chemical reaction where two or more simple reactants combine to form a single, more complex product (e.g., $A + B \rightarrow AB$) is what type of reaction?
(a) decomposition (b) single displacement (SD) (c) double displacement (DD)
(d) combination (synthesis) reaction (e) combustion reaction
4. What is molarity of aqueous solution in which 0.50 moles KCl dissolved to final 500.0 mL volume?
(a) 0.25 M (b) 0.50 M (c) 1.0 M (d) 2.0 M (e) 0.0010 M $\frac{0.50 \text{ mol} \times 1000 \text{ mL}}{500.0 \text{ mL}}$
5. If you take 10.0 mL of a 2.5 M stock solution of NaCl and dilute it with water to a final volume of 50.0 mL, what is the final concentration (C_2)?
(a) 0.10 M (b) 0.50 M (c) 1.25 M (d) 5.0 M (e) 0.050 M
6. A solution is prepared by mixing 10.0 g of sugar (solute) with 90.0 g of water (solvent). What is the percent by mass of the sugar in the solution?
(a) 1.0% (b) 10.0% (c) 11.1% (d) 90.0% (e) 100.0%
7. Calculate the **pressure** of $n = 2.00$ mol of gas at $T = 273$ K, $V = 44.8$ L. ($R = 0.0821$ L·atm/[mol·K])
(a) 1.00 atm (b) 2.00 atm (c) 0.500 atm (d) 22.4 atm (e) none of the above
8. Convert a **pressure** of 760 mmHg to atm.
(a) 0.500 atm (b) 1.00 atm (c) 7.60 atm (d) 76.0 atm $760 \times \frac{1}{760}$
9. Which law states that **pressure** and **volume** are inversely related, that is $P_1V_1 = P_2V_2$?
(a) Boyle's Law (b) Charles' Law (c) Gay-Lussac's Law (d) Avogadro's Law
10. Which of the following is a **combustion** reaction?
(a) $\text{CH}_4 + 2 \text{O}_2 \rightarrow \text{CO}_2 + 2 \text{H}_2\text{O}$ (b) $\text{NaCl} + \text{AgNO}_3 \rightarrow \text{AgCl} + \text{NaNO}_3$
(c) $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$ (d) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
(e) none of the above
11. Which of the following is the correct oxidation number of **hydrogen** in H_2O molecule?
(a) 0 (b) -1 (c) +1 (d) +2 (e) none of the above
12. A substance that completely dissociates into ions when dissolved in an aqueous solution and conducts electricity strongly is classified as a:
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 (a) gas molecules exert significant attractive forces on each other
 (b) the molecules that make up the gas phase must interact with their liquid phase
 (c) gas collisions are completely elastic (energy is not lost in collision)
 (d) the pressure of the gas is inversely proportional to temperature *2-105*
17. Which of the following salts is generally considered **soluble** in water?
 (a) AgCl (b) PbI_2 (c) BaSO_4 (d) KNO_3 (e) CaCO_3
18. A solution that contains the maximum amount of dissolved solute possible at a given temperature is called a:
 (a) unsaturated solution (b) saturated solution (c) supersaturated solution
 (d) dilute solution (e) concentrated solution
19. A gas has $P_1 = 2.00$ atm, $V_1 = 4.00$ L. If pressure increases to $P_2 = 8.00$ atm, what is V_2 ? *$P_1 V_1 = P_2 V_2$
 $V_2 = \frac{P_1 V_1}{P_2}$*
 (a) 0.50 L (b) 1.00 L (c) 2.00 L (d) 16.0 L (e) none of the above
20. Hydrogen bonding can only occur between molecules where hydrogen is directly bonded to one of three highly electronegative atoms. These atoms are:
 (a) C, H, O (b) S, Cl, Br (c) Cl, Br, I (d) N, O, F (e) Na, K, Li
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 (a) Na^+ (b) Cl^- (c) Ag^+ (d) NO_3^- (e) both (a) and (d)
24. Which statement best describes "like dissolves like"?
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25. A precipitate forms and you have to write the chemical reaction equation. How will you indicate the compound that precipitated?
 (a) (s) (b) (l) (c) (g) (d) (aq) (e) none of these

Circle either **True** or **False** for the following statements:

26. A solution that conducts electricity must contain ions.

True False

27. Combustion reactions always involve oxygen as a product.

True False

28. In a balanced chemical equation, the number and type of atoms must be equal on both sides.

True False

29. The oxidation number of a pure element is always zero.

True False

30. In a chemical reaction equation, compounds to the right of the reaction arrow are called **reactants**.

True False

31. The oxidation number of any Group 1 metal (like Na or K) in any compound is always +1.

True False

32. The total pressure of a gas mixture is equal to the sum of the partial pressures of all the individual gases in the mixture.

True False

33. The solid that forms from a solution during a chemical reaction, often by mixing two solutions, is called a **precipitate**.

True False

34. At STP, the temperature is 25 °C and pressure is 1 atm.

True False

35. The variable R in the Ideal Gas Law equation ($PV=nRT$) is known as the **universal gas constant**.

True False

"Ideal gas Constant"

$$PV = nRT$$

$$P = \frac{(200)(0.0227)(273)}{44.8}$$

$$(200)(273)(44.88)$$

$$\frac{C_1}{V_1} = \frac{C_2}{V_2}$$

$$C_2 = V_2 \times \frac{C_1}{V_1}$$
$$50.0 \text{ mL} \times \frac{2.5 \text{ M}}{10.0 \text{ mL}}$$

$$C_2 =$$

$$12.5$$